H&M; FASHION INTELLIGENCE

A plain-English guide to the data, dashboard, models, deployment and honest limitations of the H&M; analytics project.

PUBLIC DASHBOARD
https://hm-fashion-intelligence.streamlit.app/

GitHub: github.com/HuzaifaQuaid/hm-fashion-intelligence

Built from the H&M; Personalized Fashion Recommendations dataset. The data covers September 2018 to September 2020. Prices are anonymised and are not shown as currency.

01 / START HERE

Project in one minute

This project turns H&M; purchase-history data into an interactive fashion-retail intelligence app. It combines transactions, customers and article attributes into validated DuckDB tables, then produces business summaries, forecasts, recommendations, a repurchase model and a grounded analyst.

The app is historical, not live. It does not connect to H&M; systems, accept new transactions or update inventory. It is an educational and portfolio analytics product that emphasizes honest definitions and evaluation.

What it can answer

- Which departments, colours and articles are most frequently purchased?
- Is online or in-store purchasing larger, and how does online share change over time?
- How many customers return, and how do first-observed cohorts retain over later months?
- What purchase-event volume might the next month have based on historical backtests?
- Which recommendation method performs better on a temporal holdout?

What it cannot answer

- Profit or margin: product costs are absent and prices are anonymised.
- Delivery, reviews, sellers, payments or freight: those fields do not exist.
- True order value or basket composition: there is no order id.

Central rule
A number is useful only when its row meaning, denominator, time window and limitations are clear.

02 / RUN IT

Local, real-data and public use

Open the public app

https://hm-fashion-intelligence.streamlit.app/

Visitors only need a browser. Streamlit Community Cloud reads the public GitHub repository and starts app.py. It uses a compact approximately 5.8 MB analytical snapshot—not the raw 3.2 GB transaction file.

Run locally

Rebuild from the real H&M; files

The raw CSVs must be downloaded under the H&M;/Kaggle competition terms and placed in data/raw/. Raw files are ignored by Git and are not uploaded to GitHub.

If the app shows an old version
Refresh after Streamlit Cloud finishes redeploying. The public app is connected to the master branch of the GitHub repository.

03 / FOUNDATION

From CSVs to trustworthy tables

The source contains 31,788,324 raw transaction rows before exact duplicate removal. The pipeline reads fixed column types, checks keys and relationships, then writes compact analytical tables.

Table	One row means	Purpose
Transactions	One customer buying one article on one day	Date, article, anonymised price and channel.
Customers	One customer record	Age, club status and newsletter fields.
Articles	One catalogue article	Department, garment group, colour and product attributes.
Fact transactions	One validated purchase event	Foreign-key-safe analytical fact table.
Fact visits	One customer on one date	Explicit basket/visit proxy with units and scaled value.
App tables	Aggregates and article summaries	Small tables read by Streamlit Cloud.

Quality observations

- 2,974,905 exact duplicate transaction rows were removed. Because there is no quantity column, some duplicates may have represented multiple units.
- 15,861 customer records have missing age.
- 9,699 customers have no matching purchase in the transaction table.
- 995 articles are never sold in the observed history.
- Conflicting keys, orphan references, negative prices and unit-conservation failures stop the build.

Why the public app is small
The full fact table stays local. The public snapshot contains aggregate dashboard tables, article summaries, reports and model artifacts, keeping memory usage suitable for Streamlit Cloud.

04 / DASHBOARD

Workspace tour

Workspace	What it shows
Business overview	Purchase events, customers, visits, online share, department and colour leaders, monthly trends.
Channel	Online versus in-store purchase events and online-share trend.
Customers	Repeat customers, age bands, first-observed cohorts and retention heatmap.
Repurchase model	90-day repeat-visit target, ROC-AUC, precision, lift and calibration.
Forecasting	Next-month purchase-event forecast and rolling-origin comparison.
Recommendations	Similar articles and popular articles, evaluated temporally.
AI analyst	Plain-language questions routed to approved evidence tools with caveats.
Data quality	Readable validation observations and source details.

How to read it

- Purchase events are rows after exact duplicates are removed—not confirmed customer orders.
- Visits are customer-date combinations. Five articles bought on one day count as one visit with five units.
- H&M; prices are scaled/anonymised. Treat them as relative values, not dollars or other currency.
- Cohort dates mean first observed month, not guaranteed first-ever purchase.

Evaluation before claims

Repurchase model

The model asks whether a customer makes another purchase visit within 90 days of their first observed visit. Recent customers without a full follow-up are excluded. Older customers train the model; later customers are held out.

Metric	Result	Meaning
Base repeat rate	47.1%	Share of matured customers with another visit.
ROC-AUC	0.56	Ranking ability; 0.50 is chance.
Top-decile lift	1.17x	Top scores compared with the overall test rate.

Interpretation: the model is weak. It is a historical ranking signal, not a promise about a customer.

Forecasting

The target is monthly purchase events. There are 23 complete months. Methods are compared through rolling-origin backtests. The selected method is **seasonal naive** because it has the lowest backtest RMSE.

Recommendations

On the temporal holdout, popularity MAP@10 is 0.0034; content-based MAP@10 is 0.0024. Popularity wins this split, so the app does not claim personalization is better.

06 / ANALYST

Evidence, not invented answers

The AI analyst is not a free-form database chatbot. It has a fixed whitelist of approved tools: sales overview, channel split, department movement, customer overview, forecast, recommendations, repurchase and data quality.

- A deterministic router, or an optional language model, chooses one approved tool.
- The tool reads known project tables or saved reports.
- The response includes an answer, evidence and limitations.
- Unsupported questions about profit, inventory, delivery, reviews, returns, income and marketing are refused.
- It does not generate arbitrary SQL from the user question.

Example
“Is online or in-store bigger?” returns channel counts and percentages. “What was the delivery time?” explains that shipping data is absent instead of inventing a number.

Current results

Measure	Result
Purchase events	28,813,419
Customers	1,362,281
Repeat customers	67.1%
Average visits per customer	6.67
Median age	32
Online share	31.7%

Suggested project description

I built an evidence-first H&M; fashion-retail intelligence app from 28.8 million purchase events. The pipeline validates three source tables, removes exact duplicates, creates DuckDB marts and ships only compact aggregates to Streamlit Cloud. The dashboard covers channel behaviour, cohorts, repeat purchasing, forecasts and recommendations. I report the limits honestly: prices are anonymised, there is no order id, the repurchase model is weak, popularity beats content-based recommendations on the holdout, and the forecast selects a simple baseline.

08 / GLOSSARY

Plain-English technical terms

Term	Plain-English meaning
Anonymised price	A transformed price that cannot be interpreted as a real currency amount.
Backtest	A historical simulation that trains using earlier data and tests against later data.
Calibration	Whether predicted probabilities match observed frequencies.
Cohort	A group sharing a starting point; here, first observed purchase month.
DuckDB	A compact analytical database that runs locally without a separate server.
Feature	An input variable supplied to a model.
Leakage	Giving a model information that would not be known at prediction time.
MAP@10	A recommendation score rewarding relevant articles near the top ten positions.
ROC-AUC	A ranking score; 0.50 is chance and 1.00 is perfect.
Temporal split	Separating training and test observations by date.
Holdout	Data kept out of model fitting for independent evaluation.
Relative value	A number useful for comparisons, but not a real money amount.
Streamlit Community Cloud	A service that runs a Streamlit Python app from GitHub and gives it a public URL.
Whitelisted tool	An explicitly approved analysis action; the analyst cannot run arbitrary SQL.

Final takeaway
This is a complete historical retail analytics MVP: validated data, compact deployment, interactive dashboard, evaluated models and a grounded analyst. Its strongest quality is not claiming more than the H&M; data can support.